Lab 8 - Bed Adhesion - Skirt vs Brim vs Raft

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 02: Process Related Issues for AM TSC objective: A2 - analyse first-layer adhesion issues and select the correct adhesion aid

Goal

Slice three instances of the same cube with a skirt, a brim and a raft, print them, and compare adhesion strength, removal effort and bottom-surface finish.

What you'll build

Three prints of cube10.stl demonstrating skirt, brim and raft - and a rule for when to use each.

Tools: Ultimaker Cura, cube10.stl, FDM printer

Steps

1. Open cube10.stl in Cura and duplicate it twice so three cubes sit on the plate.

```
Right-click > Multiply Selected - 2 copies
```

2. Select cube 1 and set Build Plate Adhesion to Skirt - a loop that primes the nozzle without touching the part.

```
Per Model Settings > Build Plate Adhesion > Skirt
```

3. Select cube 2 and set adhesion to Brim, 8 mm - a flat border fused to the part's first layer.

```
Build Plate Adhesion > Brim - width 8 mm
```

4. Select cube 3 and set adhesion to Raft - a printed platform the part sits on.

```
Build Plate Adhesion > Raft
```

5. Slice and compare the three toolpaths in Preview; note the extra time and material the raft adds.

```
Slice > Preview
```

6. Print all three cubes in one job.

```
Save to Disk > print
```

7. Remove each cube: note how the skirt never touches, the brim peels off the edge, and the raft must be pried from the base.

```
Deburring tool / scraper
```

Evidence to capture

- Cura Preview screenshot showing the skirt, 8 mm brim and raft toolpaths.
- Recorded time/material estimate for each adhesion option.

- Photo or observation table comparing removal effort and bottom-surface finish.

Test it

You can state when each aid applies - skirt to prime the nozzle, brim for warping-prone or small-footprint parts, raft for top-heavy or poorly adhering parts - and show the bottom-finish difference.

Files in this folder

- cube10.stl - Core sample model used by the lab procedure.